An Olympiad Notebook

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Preface

The Philosophy

This is a collection of notes written by me on all of the four subjects included in mainstream olympiads. Much of the contents is sourced from a various plethora of free and paid books.

This is not meant to be a one-spot go to for all your Olympiad preparation, even though I have tried to include every bit of information I could get my hands on. One important part that is missing from it are problems. The Primary reason for that being my laziness and the fact and these are meant to be my *notes*, which shall capture my Olympiad journey. This is not an olympiad go-to book. Be aware

Acknowledgments

First and foremost, my greatest thanks goes to Evan Chen for his awesome style file, parts of which I have modified and used in my notes. He has also been a source of much inspiration, and for the EGMO [Che16] book, upon which much of the Geometry is based.

I also must thank Aditya Khurmi for his freely available book MONT [Khu20], Although he did not have any particular ivolvement here, I must also thank Kevin Zhou, for inspiring me.

Notation

- $\mathbb{N} = \{1, 2, 3 \ldots\}$
- $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$

To her, whose death still haunts me.

PART A

Number Theory

Part A: Contents

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1 Divisibility

Motivation: Divisibility forms the basis of all of Number Theory.

§1.1 Basic Properties

The idea of divisibility is something most people are familiar with. Yet, it is a notion so useful, that we may yet again acquaint ourselves with it and its properties. This time, however, we are going to approach it with formality and use its property in a general, and abstract manner. For this, let us first define division for the integers.

If you are unfamiliar with these symbols, have a look at Appendix A

Definition 1.1.1 (Division)

The operation division is defined as:

1. An integer y is said to be a multiple of x if
$$y \in \{kx : k \in \mathbb{Z}\}$$
2. If y is a multiple of x then, y is *divisible* by x and x divides y which is written as:
$$x \mid y.$$

Immediately, we can put forth some *propositions* from these definitions. They may not be obvious from first glance but they are indeed very easy to get an intuitive grasp on after we have seen them.

Proposition. For x, y, $z \in \mathbb{Z}$

1. $\forall x \neq 0$, $x \mid x$ and $x \mid 0$.
2. $1 \mid x$.
3. $x \mid y$ and $y \mid z \implies x \mid z$.
4. $x \mid y \iff \dfrac{y}{x} \in \mathbb{Z}$

Most of the proofs here are trivial, but let us show that using the last proposition we can prove the third one.

Proof. $x \mid y \iff \dfrac{y}{x} \in \mathbb{Z}$ and $y \mid z \iff \dfrac{z}{y} \in \mathbb{Z}$. But $\dfrac{z}{x} = \dfrac{y}{x} \cdot \dfrac{z}{y} \in \mathbb{Z} \iff x \mid z$. □

Let $x \mid y$. Since y is a multiple of x, it might seem to be the case that $y \geqslant x$. But is that really true? Well, not really. However, there is something along those lines that is.

Theorem 1.1.2

If $x \mid y$, then either $|y| \geqslant |x|$ or $y = 0$.

▶ The 0 case is extremely important and often hard to see. Remember to look out for this case.

Proof. Since $x \mid y$, by the definition of division, we must have,

$$y = kx : k \in \mathbb{Z}$$

Since the case of 0 has already been established, let us concern ourselves with $k \in \mathbb{Z} \setminus 0$, Thus, $|y| = |k \cdot x| = |k| \cdot |x|$. But $|k| \cdot |x| \geqslant |x| \iff |y| \geqslant |x|$. □

The following is not of a severe importance in the view that it is hard to see or prove. But it allows us to quickly formalise our solutions and is thus useful for proofs.

Lemma 1.1.3

If $x \mid a$, then $x \mid ac + b \iff x \mid b$.

Proof. The proof is trivial as, $x \mid a \iff a = kx : k \in \mathbb{Z}$.

Direction 1 Note, $x \mid ac + b \iff ac + b = nx : n \in \mathbb{Z}$, hence, $b = nx - ac = nx - kcx = (n - kc)x \iff x \mid b$.

Direction 2 And, $x \mid b \iff b = mx : m \in \mathbb{Z}$, hence, $ac + b = kcx + mx = (kc + m)x \iff x \mid ac + b$. □

This proof kind of sheds light on how to prove statement of the type $P \iff Q$, or in words P holds if and only if Q does. We show that $P \implies Q$ and $Q \implies P$. You can verify that this is the same $P \iff Q$. Now, let us end this section with some properties of division and examples.

Theorem 1.1.4

Let $x, y, z \in \mathbb{Z}$

1. $\forall x \neq 0$, $x \mid x$ and $x \mid 0$.
2. $1 \mid x$.
3. $x \mid y$ and $y \mid z \implies x \mid z$.
4. $x \mid y \iff \dfrac{y}{x} \in \mathbb{Z}$.
5. $\forall z \neq 0$, $x \mid y \iff xz \mid yz$.
6. $x \mid a$, $x \mid b \implies x \mid an + bm$.
7. $x \mid y$, $y \mid x \iff y = \pm x$.
8. $x \mid y \implies x \mid yz$.

Example 1.1.5

Show that if $n \in \mathbb{Z}, n > 1$, then $n \nmid 2n^2 + 3n + 1$.

Proof. Here we can see the use of Lemma 1.1.3. Suppose $n \mid 2n^2+3n+2$, then $n \mid 2n^2+3n$, and $n \mid 2n^2 + 3n + 1 \iff n \mid 1$. But, $n \mid 1 \implies |1| \geqslant |n| \iff 1 \geqslant n$ which contradicts our earlier statement that $n > 1$. Hence, $n \nmid 2n^2 + 3n + 1$. □

Remark 1.1.6. $|1| \geqslant |n| \iff 1 \geqslant n$ because $n > 1 > 0$.

Example 1.1.7

Show that for any two natural numbers, a, $b : a > b$, $a \nmid 2a + b$.

Proof. Again, we can use Lemma 1.1.3. Suppose $a \mid 2a+b$, then $a \mid 2a$ and $a \mid 2a+b \iff a \mid b$. But, $a \mid b \implies |b| \geqslant |a|$ and $a, b \in \mathbb{N} \implies b \geqslant a$. This contradicts our the fact $a > b$. Hence $a \nmid 2a + b$. □

§1.2 Euclid's Division Lemma

Euclid's division lemma is something we will make much use of, especially later. For now we shall be satisfied with the unproven statement.

Theorem 1.2.1

For any integers a, b, we can find *unique* integers q, r, such that:

$$b = aq + r, \quad 0 \leqslant r < a.$$

Here, r is called the remainder, and q is called the quotient.

▶ By unique we really mean one and only one in the language of mathematics. This means that there has to be at least one and at most one such thing.

§1.3 Primes

Definition 1.3.1 (Primes)

Any $x \in \mathbb{N}$ is called a prime, if and only if it has exactly 2 *divisors*.

▶ Note that there have to be exactly two factors. Therefore, 1 which has only one factor is not prime.

Primes are a very interesting subset of naturals. There properties will be explored later. For now, we use primes to define one of the most important theorem in all of number theory.

§1.3.1 Fundamental Theorem of Arithmetic

Theorem 1.3.2 (Fundamental Theorem of Arithmetic)

Any natural number can be *uniquely* expressed as a product of primes upto order.

The order in which the product is expressed is irrelevant because of the commutativity of multiplication.

We can further extend this theorem to include all integers by using some claims.

§1.3.2 Integers as Multisets

Claim 1.3.3 — Any non-zero $x \in \mathbb{Z}$ can be expressed a multiset of primes(and -1). If the number is positive, then we can simply use Theorem 1.3.2 and included all the prime factors in its multiset. If the number is negative, then we express it as the multiset of its absolute value and include -1 in the multiset.

Remark 1.3.4. A *multiset* is a set where repeating elements are counted seperately. Each element has a *mulitplicity* which indicates the number of times that element appears in the multiset.

We could, for example, express 20 as $\{2, 2, 5\}$, and -20 as $\{-1, 2, 2, 5\}$.
This allows us to restate Theorem 1.3.2.

Theorem 1.3.5 (Fundamental theorem of Arithmetic in multisets)

Any natural number can expressed as a multiset of primes.

Remark 1.3.6. This multiset, is, by definition *unique*.

It also allows us to restate divisibility for positive integers.

Theorem 1.3.7 (Divisibility in multisets)

$\forall a, b \in \mathbb{N}$,

$$a \mid b \iff A \subseteq B$$

Where A and B are multisets of a and b respectively. We can't extend this to the negative integers because of the rather annoying presence of -1.

Remark 1.3.8. Henceforth, all variables are integers unless explicitly mentioned.

§1.4 GCD and LCM

§1.4.1 GCD

Definition 1.4.1

$\gcd(a, b)$ is the multiset of all *common* prime factors of a and b.

This gives us the following properties of gcd.

Proposition 1.4.2 —

- $\gcd(a, b)$ is the greatest integer which divides both a and b. In particular, $\gcd(a, b) \leqslant a, b$.
- $c \mid a,\ c \mid b \iff c \mid \gcd(a, b)$.

We can, in fact, show that the two definition of gcd are equivalent.

Proof. Let $\mathcal{S} = \{p : p \mid a, b \text{ and } p \text{ is a prime}\}$. Suppose that $\gcd(a, b) \neq \mathcal{S}$. Then $\gcd(a, b) > \mathcal{S}$.

Note that $\gcd(a, b)$ must contain an element $x \notin \mathcal{S}$ where x is prime.(Since it is an element in the multiset of $\gcd(a, b)$).

But, $x \in \gcd(a, b) \iff x \mid \gcd(a, b)$, and $x \mid \gcd(a, b)$, $\gcd(a, b) \mid a, b \implies x \mid a, b$. x is prime, and $x \mid a, b \implies x \in \mathcal{S}$. This contradicts our earlier deduction that $x \notin \mathcal{S}$. By *reductio ad absurdum*, we have $\gcd(a, b) = \mathcal{S}$. □

Claim 1.4.3 — Let $a = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_n^{\alpha_n}$, $b = p_1^{\beta_1} p_2^{\beta_2} \dots p_n^{\beta_n}$. Then,

$$\gcd(a, b) = p_1^{\min(\beta_1, \alpha_1)} p_2^{\min(\beta_2, \alpha_2)} \dots p_n^{\min(\beta_n, \alpha_n)}$$

Where, $\alpha_i, \beta_i \in \mathbb{N}_0$.

This is simply a consequence of the definition of gcd.

Definition 1.4.4 (GCD)

We may now state that the following are equivalent.

- $\gcd(a, b) = A \cap B$
- $\gcd(a, b)$ is the greatest integer which divides both a and b. In particular, $\gcd(a, b) \leqslant a, b$.
- Let $a = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_n^{\alpha_n}$, $b = p_1^{\beta_1} p_2^{\beta_2} \dots p_n^{\beta_n}$. Then,

  $$\gcd(a, b) = p_1^{\min(\beta_1, \alpha_1)} p_2^{\min(\beta_2, \alpha_2)} \dots p_n^{\min(\beta_n, \alpha_n)}$$

  Where, $\alpha_i, \beta_i \in \mathbb{N}_0$.

§1.4.2 LCM

Definition 1.4.5 (LCM)

The following three definitions are equivalent.

- $\operatorname{lcm}(a, b) = A \cup B$
- $\operatorname{lcm}(a, b)$ is the least number divisible by both a and b. Particularly, $\operatorname{lcm}(a, b) \geqslant a, b$.
- Let $a = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_n^{\alpha_n}$, $b = p_1^{\beta_1} p_2^{\beta_2} \dots p_n^{\beta_n}$. Then,

  $$\operatorname{lcm}(a, b) = p_1^{\max(\beta_1, \alpha_1)} p_2^{\max(\beta_2, \alpha_2)} \dots p_n^{\max(\beta_n, \alpha_n)}$$

From the definitions of gcd and lcm we obtain the following very important theorem.

Theorem 1.4.6

$$\operatorname{lcm}(a, b) \cdot \gcd(a, b) = a \cdot b$$

Proof.

$$\begin{aligned}\gcd(a,b)\cdot\operatorname{lcm}(a,b) &= p_1^{\min(\alpha_1,\beta_1)+max(\alpha_1,\beta_1)}\dots p_n^{(\max(\alpha_n,\beta_n)+\min(\alpha_n,\beta_n))}\\ &= p_1^{\alpha_1+\beta_1}\dots p_2^{\alpha_n+\beta_n}\\ &= a\cdot b\end{aligned}$$

□

§1.5 Co-prime

Definition 1.5.1

Two numbers, a, b are called co-prime if $\gcd(a,b)=1$.

Definition 1.5.2

A number x is pairwise co-prime to n other numbers $p_1, p_2\dots p_n$, if $\gcd(x,p_1)=\gcd(x,p_2)\dots=\gcd(x,p_n)=1$.

§1.6 Infinitude of Primes

Theorem 1.6.1 (Euclid)

There are infinitely many primes.

Proof. Let there be finitely many primes, $\{p_1,p_2\dots,p_n\}$. Define N as $p_1p_2\cdots p_n+1$.

Note that N is pairwise co-prime to all, $p_1p_2\cdots p_n$ because if $p_i\in\{p_1,p_2\dots,p_n\}\mid N$ and $p_i\mid p_1p_2\cdots p_n \iff p_i\mid 1$. But, $p_i\in\mathbb{N}\implies p_i=1$ which is absurd.

Since $N\in\mathbb{N}$, N must have a prime factor, p because of Theorem 1.3.2. And $p\notin\{p_1,p_2\dots,p_n\}$ because N is pairwise co-prime to all primes in $\{p_1,p_2\dots,p_n\}$. This contradicts our assumption that $\{p_1,p_2\dots,p_n\}$ was the set of all primes, proving that there are infinitely many primes.

□

This type of proof is called the method of contradiction. We assume the negation of what we want to prove, say to prove P, assume $\neg P$ and then show that the negation leads to a contradiction. Thus, $\neg P$ cannot hold, and therefore P must hold.

§1.7 Euclid's Division Algorithm

Euclid's division algorithm is very useful in a lot of cases. It allows to compute $\gcd(a,b)$ faster, and can be helpful in some proofs as well. The basic idea of it is that since we can represent $a=bq+r$ for any a and b, We can substitute this as a while computing $\gcd(a,b)$.

Theorem 1.7.1 (Euclid's Division Algorithm)

For any two integers a, b, such that $a=bq+r$, where $0\leqslant r<a$,

$$\gcd(a,b)=\gcd(r,b)$$

Proof. The idea here is to notice that $\gcd(a, b) \mid a$ and $\gcd(a, b) \mid b$. Therefore, since $\gcd(a, b) \mid a$,

$$\gcd(a, b) \mid bq + r$$

But, $\gcd(a, b) \mid b$. Citing Lemma 1.1.3, we must have $\gcd(a, b) \mid r$. Now that we have shown $\gcd(a, b) \mid r, b$, we must also show that it is the greatest number to do so. Let us assume that is not the case, let $\ell \mid r, b : \ell > \gcd(a, b)$. Which means that $\ell = \gcd(r, b)$

But the thing of note here is that if $\ell \mid r$ and $\ell \mid b$. Then $\ell \mid a$. But that can not happen, since ℓ is greater than $\gcd(a, b)$ and divides both a, b. Therefore, $\ell = \gcd(a, b) = \gcd(r, b)$. □

One thing to note here, is that since $\gcd(a, b) = \gcd(r, b)$, $\gcd(a, b) = \gcd(a - bq, r)$.

Basically, $\gcd(a, b) = \gcd(a \pm kb, r)$**.**

This statement is equally, if not more important than the formal one.

While it is true that $\gcd(a, b) = \gcd(r, b)$**, it is not in any way true that** $\operatorname{lcm}(a, b) = \operatorname{lcm}(r, b)$**.** !

§1.8 Bezout's Lemma

We are now nearing the end of this chapter, and bezout's lemma is one of the last topics we'll cover. Consider the values of $ax + by$, for $a, b, x, y \in \mathbb{Z}$. For some constant a and b, say 2 and 4, you may notice that the $\gcd(a, b)$ is the smallest non-negative value of $ax + by$!

Is this always true? In this case, yes. Although I will not go much into the motivation, it is an extremely useful tool.

Theorem 1.8.1 (Bezout's Lemma)

For two integers a and b, we can always find integers x and y such that,

$$ax + by = \gcd(a, b)$$

add proof

§1.9 Base systems

Whenever we write something like 28 what we really mean is 28 in base 10 or $28_{(10)}$. To represent it in an expanded form, we may write something like,

$$28 = 8 \times 10^0 + 2 \times 10^1$$

In general, for a number n in base ℓ, we can write something like,

$$n = a_0 \times \ell^0 + a_1 \times \ell^1 + \cdots \times a_k \times \ell^k$$

Where $0 \leqslant a_0, a_1, \ldots, a_k < \ell$. This representation is unique. If there is another such representation in the same base, it must violate the conditions established. Note that $a_k \neq 0$, because if it is, then we can simply reduce to $k - 1$ and rewrite similarly. Thus $n \geqslant \ell^k$. But n $n < \ell^{k+1}$ because otherwise the representation would be not be unique. If n is represent in such a manner, then it is written with digits $(a_k a_{k-1} \ldots a_1)_{(\ell)}$.

Also similarly, you may notice that $a\ell^k \leqslant n < (a + 1)\ell^k$. These help us devise an algorithm to represent n in base ℓ.

2 Basics of Modular Arithmetic

Motivation: Modular arithmetic allows us to expand the notion of equality.

§2.1 "Modulo"

It is very hard to and rarely needed to establish equality in number theory. What may do instead is the build on the notion of *modular* arithmetic. Consider an analog clock, if it starts at any arbritary time, then no matter how many rotations it goes through, whenever it points at 1, the time is 1 a.m or p.m. Thus even if we start counting hours, we may note that $13 = 1 = 25 = \dots$ The equality symbol is not apt here. What should be much better used is the symbol of equivalence, "$\equiv$". Thus, what we mean is $13 \equiv 1 \pmod{n}$. In general,

Definition 2.1.1 (Congruent)

$a \equiv b \pmod{n}$ if and only if $p \mid a - b$. In such a case a is said to be congruent to b modulo n.

A neat little property of this is that $a + n \equiv a \pmod{n}$. Also, we may note that the values of b such that $b \equiv a \pmod{n}$ form an A.P.

Proof. $b \equiv a \pmod{n} \iff n \mid b - a \iff b - a = nk$ where k is an integer. Thus, $b = a + nk$, which is an A.P. □

§2.2 Remainder and Modulo

Consider the set of values for which $b \equiv 3 \pmod{5}$. They form the set, $\{\cdots - 7, -2, 3, 8, \dots\}$. It is difficult to work with such infinite sets which expand in both directions, so let us concern ourselves only with non-negative elements, $\{3, 8, 13, 18 \dots\}$.

Through this example, we may see a neat pattern,

Lemma 2.2.1

The lowest non-negative value of a such that $a \equiv b \pmod{n}$ is the remainder r such that $b = nk + r$, $0 \leqslant r < n$.

§2.3 Residue class

Consider a number 3. If we concern ourselves only with the remainder such that $a \equiv r \pmod{n}$, you may note that r can only have three values here, 0, 1, and 2. Thus, either $a \equiv 0$, $a \equiv 1$ or

$a \equiv 2 \pmod{n}$. This is always true. These 3 sets that a can belong to, are called the 3 residue classes modulo 3.

In general,

Definition 2.3.1

For any natural number n, r such that $0 < r < n$, the rth residue class is the set of all integers a such that $a \equiv r \pmod{n}$. This set is,

$$\{\dots, r-2n, r-n, r, r+n, r+2n, \dots\}$$

There are exactly n residue classes modulo n since r can take n values, from 0 to $n-1$.

§2.4 Arithmetic over modulo

Consider $a+b \pmod{n}$ for $a \equiv r \pmod{n}$ and $b = m \pmod{n}$. Does $a+b \equiv r+m \pmod{n}$? Yes. The idea behind the proof is to just have $a = nk + r$ and $b = nd + m$.

Proposition 2.4.1 — From the definition and the previous discussion of modular arithmetic, we have,

1. For $a \equiv r$, $b \equiv m \pmod{n}$, $a+b \equiv r+m \pmod{n}$
2. $a \equiv r$, $b \equiv m \pmod{n} \implies ab \equiv rm \pmod{n}$
3. For $a \equiv b \pmod{n}$, $a+c \equiv b+c \pmod{n}$
4. For $a \equiv b \pmod{n}$, $ac \equiv bc \pmod{n}$
5. For $a \equiv b \pmod{n}$, $a^c \equiv b^c \pmod{n}$

PART B

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